

EXPERIMENT – 1 INTERFACING WITH ARDUINO

PROJECT NAME

Temperature Monitoring Alert System

OBJECTIVE / PROBLEM STATEMENT

This project monitors temperature using a DHT11 sensor and activates a buzzer when the temperature exceeds the threshold value of 50°C. The system is designed using Arduino UNO in SimulIDE.

COMPONENTS USED

Component	Quantity
Arduino UNO	1
DHT11 Temperature Sensor	1
Buzzer	1
Jumper Wires	Required

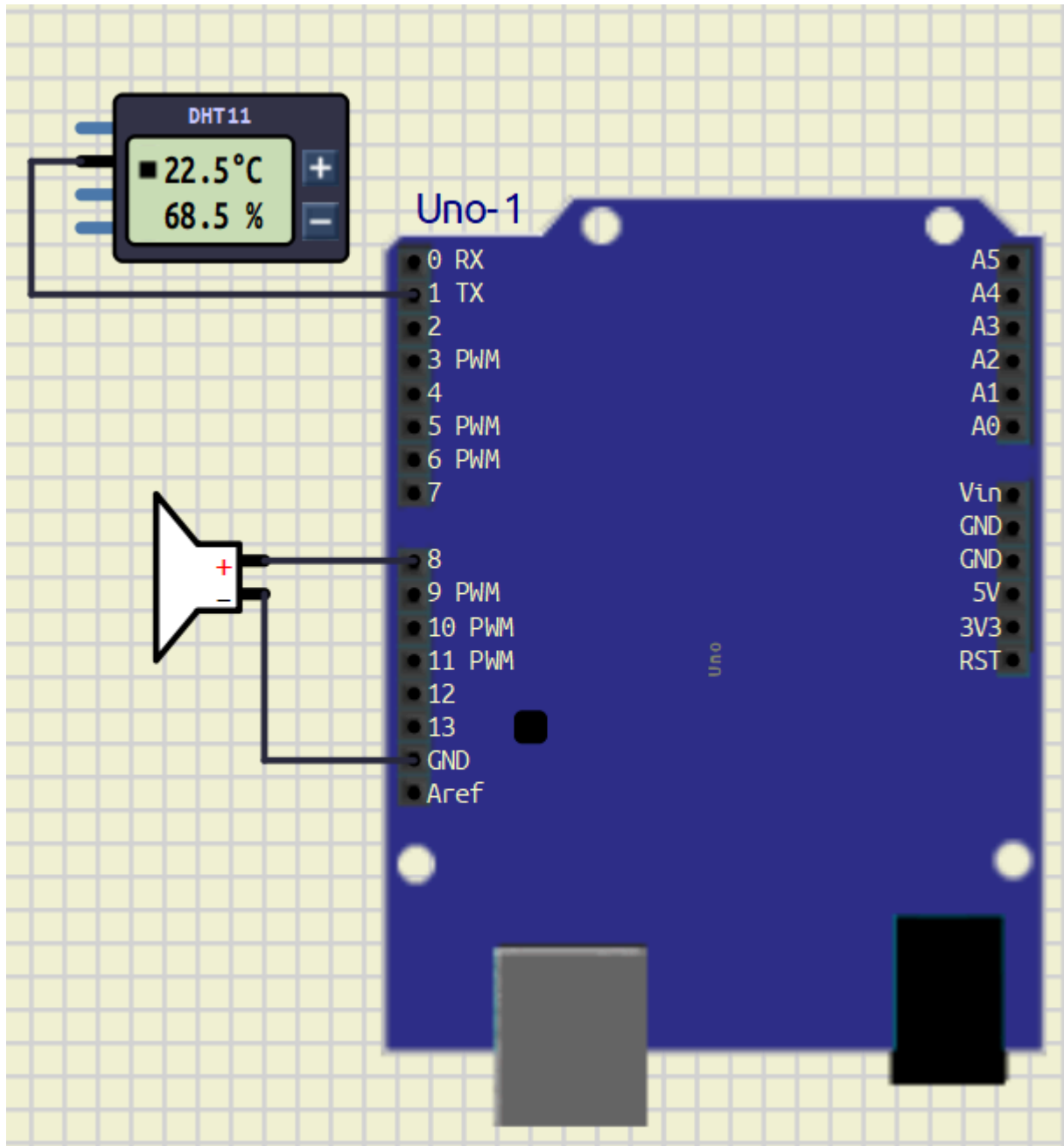
PIN CONNECTIONS

Device	Arduino Pin
DHT11 Data Pin	D1
Buzzer	D8
DHT11 VCC	5V
DHT11 GND	GND

SOFTWARE USED

- Arduino IDE
 - SimulIDE
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Circuit diagram



REQUIRED LIBRARY

```
#include <DHT.h>
```

MAIN ARDUINO CODE

Code

```
#include <DHT.h>

#define DHTPIN 1
#define DHTTYPE DHT11
#define BUZZER 8
#define THRESHOLD 50
```

```
DHT dht(DHTPIN, DHTTYPE);

void setup()
{
    pinMode(BUZZER, OUTPUT);
    dht.begin();
}

void loop()
{
    float temp = dht.readTemperature();

    if(temp >= THRESHOLD)
    {
        digitalWrite(BUZZER, HIGH);
    }
    else
    {
        digitalWrite(BUZZER, LOW);
    }

    delay(1000);
}
```

WORKING PRINCIPLE

1. DHT11 sensor continuously reads temperature.
2. Arduino UNO compares the temperature with the threshold value.
3. If temperature is greater than or equal to 50°C:
 - Buzzer turns ON.
4. If temperature is below 50°C:
 - Buzzer remains OFF.
5. The process repeats continuously.

OUTPUT

- Buzzer remains OFF below 50°C.
- Buzzer turns ON at or above 50°C.
- System performs real-time temperature monitoring.

ADVANTAGES

- Simple and low-cost system
- Easy to implement
- Real-time monitoring
- Useful for safety applications

APPLICATIONS

- Fire warning systems
 - Temperature alert systems
 - Industrial monitoring
 - Home safety projects
 - Embedded system learning
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CONCLUSION

The Digital Temperature Monitoring System using Arduino UNO and DHT11 sensor was successfully implemented in SimulIDE. The buzzer activated whenever the temperature crossed the threshold value of 50°C.

References

1. Arduino Project hub- With DHT11 sensor <https://projecthub.arduino.cc/arcaegecengiz/using-dht11-12f621>
2. DHT sensor Library - <https://docs.arduino.cc/libraries/dht-sensor-library/#Releases>
3. Arduino Uno board Pin Out - <https://deepbluembedded.com/arduino-uno-pinout/>